ADAM SOBIESZEK

Machine Learning Researcher

Contact

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Location Warsaw, Poland

Summary

Machine learning researcher with 6 years of experience developing new methods, and heading 10 national or international research teams.
As Jutro Medical's first AI specialist, built LLM agents that reduced doctors' time on AI-assisted visits by 75% and reached over 20% of all visits for 160k patients.
Co-authored 13 scientific publications in AI, Neuroscience and Psychology, including one of 2022's most read on the abilities of Large Language Models (featured also on Hacker News).
My individual research work focuses on deep learning methods for accelerating research in medicine, neuroscience and social science, now as Principal Investigator of a Ministry of Science grant on LLM agents.

Selected Work

- **LLM agents:** Conducting medical interviews and assisting doctors, used in tens of thousands of real visits
- **Brain-Computer Interfaces:** State-of-the-art deep learning models decoding imagined speech from live EEG
- **Generative models:** State-of-the-art methods for predicting and manipulating human judgments of faces
- **Language transformers:** State-of-the-art method for predicting human ratings of words (r=0.91)
- **EEG transformers:** Predicting diseases from EEG recordings, for the National Center for Research and Development
- **Python software:** Suite for large scale LLM data processing and faster generative model inference
- **All code:** github.com/AdamSobieszek

Education

2019-2023; -2025 University of Warsaw

Master's at the College of Inter-Faculty Individual Studies in Mathematics and Natural Sciences

- Masters: Quantitative Social Research, Psychology
- Bachelors: Physics and Neuroinformatics

Awards

- **2026** "Pearls of Science" (Perły Nauki) Ministry of Science 4-year Grant award of 200'000PLN for continuing my research on Large Language Models
- **2023** 'Debut of the Year' and 'Science' award, **2024** 'Society of the year' award; National Student Science Movement Awards (StRuNa) for my research work as president of the Society for Social Artificial Intelligence
- **2023, 2022** Minister of Education Scholarship for Significant Scientific Achievements
- **2020** Minister of Education Scholarship for Outstanding Scientific Achievements

Earlier: Philosophy Olympiad 2019; Math Olympiad 2015

Experience

2023-2026
Jutro Medical

Machine Learning Researcher Lead

- Before finishing my university studies I was scouted by Jutro Medical to join as **their first AI specialist**, and singlehandedly develop the firm's AI strategy
- Built **LLM-based predictive models** based on hundreds of thousands of electronic health records, using which I identified which 25% of doctor visits could benefit most from AI automation
- In the next 2 years I built **LLM agentic solutions** that conduct medical interviews with patients, write medical documentation, triage and assist doctors, that together **reduced by 75% the time doctors had to spend on AI-assisted visits** (from 12 to 3 minutes), which in 2025 helped secure the firm's **Series A funding of 151'000'000PLN**
- Working with doctors and now a dedicated AI team I helped assemble, by 2026 the adoption of the AI automation methods we developed reached **over 20% of all doctor visits** for Jutro Medical's 160k patients
- Represented Jutro Medical at AI industry events, including the London AI in Healthcare Hackathons at UCL

2025-2029
University of Warsaw

Principal Investigator of Research Grant on Scientific use of LLM Agents

- Using LLM agents as a tool in experimental psychology: **simulating groups of agents in dialogue** to find hypotheses about effective group reasoning, then testing them in three experiments, also with human-AI groups
- Developing **new modular LLM architectures** based on the interactionist theory of reasoning of Mercier and Sperber, with modules fine-tuned using LoRA and human feedback
- Building **open source Python packages** for social simulations with LLMs and for stimuli selection

2025

Inovo.vc AI Hackathon (PL.Kombinator #1)

- In a 2 person team **won 1st place and $10,000 in prize money**. Achieved **state-of-the-art performance for a Brain-Computer Interface** decoding imagined speech from EEG signals

2022-2023

National Center for Research and Development

Machine Learning Researcher

- **Led part of the public-private partnership research team** developing a transformer deep learning model for the automatic detection of neurological disorders, and built it into **a tool for physicians to assist in EEG exams**
- Developed **NLP methods** of extracting information from doctors' notes to **predict health outcomes**

2021-present

President of the Society for Social Artificial Intelligence, University of Warsaw

- **Managed several research teams of 3 to 8 people;** conducted lectures and workshops. The society had up to 50 members.

2020-present

Various institutions

Individual Research Work

- Developed a method of **using GANs for precise manipulation** of 32 psychological impressions of faces, tested on hundreds of participants. **Built it into a GUI application for researchers**, used e.g. with C. Thorsten (RIT, New York) to improve video game graphics
- Developed **state-of-the-art sentiment prediction method for words, achieving a correlation of r=0.91** with human judgments
- Proposed and headed three international research projects at Brainhack Warsaw and Cracow, which developed a new regularisation method for GANs and **reduced training time by 60%** compared to the literature
- Worked on **methods to increase LLM task performance**, in collaborations with Dragos Gorduza (Oxford Internet Institute), and Hugo Mercier (ENS Paris)

2020-2022

Faculty of Physics, University of Warsaw

Co-investigator, National Research Grant

- **Member of a neuroscience research team** that mapped out the neuronal correlates of different emotional states during language processing. My work focused on **statistical analysis, signal analysis and programming**

2019-2023

Faculty of Psychology, University of Warsaw

Research Associate and Python Programmer

- Developed an algorithm for automatic stimuli set creation that **saves up to 8h of prep. work per study**; programmed rest APIs, NLP applications and probabilistic models in research grants

Languages

Polish, English (fluent; C2 Cambridge English)

Research Interests

- **Large Language Models:** Methods to increase LLM task performance, libraries for building LLM applications, the limits of these models
- **Generative models:** GANs and flow matching for image manipulation
- **Medicine:** Deep learning for EEG signal data and NLP for doctors' notes
- **Mathematics:** Applications of Differential Geometry to Deep Learning
- **Programming:** Python (expert), PyTorch, R, Matlab (advanced), Java, JavaScript, SQL (basic)

Scientific Research

PEER-REVIEWED JOURNAL ARTICLES

2025 **Generative Neural Networks for Experimental Manipulation**
Sobieszek, A., et al. · *British Journal of Psychology*

2022 **Playing Games with AIs: The Limits of GPT-3 and Similar Large Language Models**
Sobieszek, A. & Price, T. · *Minds and Machines, 32(2), 341–364* Highlighted by Y Combinator's Hacker News

2023 **Affective Norms Extrapolation Using Transformer-based Neural Networks and Its Application to Experimental Stimuli Selection**
Sobieszek, A. & Plisiecki, H. · *Behavior Research Methods*

2024 **Emotion topology: extracting fundamental components of emotions from text using word embeddings**
Plisiecki, H. & **Sobieszek, A.** · *Frontiers in Psychology*

2022 Affect Misattribution in Ambiguous Stimuli and Its Neural Correlates Imbir, K. K., ..., **Sobieszek, A.**, et al. · *Brain Sciences, 12(8), 1093*

2021 Electrophysiological correlates of interference control... Imbir, K. K., ..., **Sobieszek, A.**, et al. · *PLOS ONE, 16(10), e0258177*

2022 Valence aligned words in the processing of an emotional categorisation task Imbir, K. K., ..., **Sobieszek, A.**, et al. · *PLOS ONE, 17(3), e0265537*

2020 Electrophysiological and Behavioral Correlates of Valence, Arousal... Imbir, K. K., ..., **Sobieszek, A.**, et al. · *Frontiers in Human Neuroscience, 14*

2023 Language and Cognition · 2022 PLOS ONE · and:

13 SCIENTIFIC PUBLICATIONS — SEE ALL PUBLICATIONS → researchgate.net/profile/Adam-Sobieszek

CONFERENCE TALKS

2026 Barcelona, Spain · **APS Annual Convention**
Psychgan: Advances In Trait Manipulation of Face Stimuli with Principles of Experimental Design and Neural Networks

2025 Washington D.C., US · **APS Annual Convention**
Theory of Optimal Experimental Manipulation for Psychological Traits of Stimuli Using Generative Networks

2025 San Diego, US · **SPSP Annual Convention**
Stimuli Pursuit: New Algorithm to Better Leverage Stimuli Sets in Experimental Psychology

2024 Warsaw, Poland · **Annual Meeting of the Polish Social Psychological Society**
Neural Network Methods of Explaining and Manipulating Warmth and Competence Judgments in Face Images

2022 Milano, Italy · **ESCON Transfer of Knowledge Conference**
Generative Neural Network Approach in Evaluation of the Dominance-Trustworthiness Model of Face Impressions

2022 Cracow, Poland · **Measuring Behavior**
Generative Neural Networks for Experimental Manipulation of Complex Psychological Impressions in Face Perception Research and Beyond

OTHER PUBLICATIONS

2023 **Analiza uniwersalności emocji przy użyciu wielojęzykowych modeli uczenia maszynowego typu transformer**
Plisiecki, H. & **Sobieszek, A.** · *Monograph chapter · TYGIEL „Interdyscyplinarność kluczem do rozwoju"*